

The Editors
Journal of Forecasting

September 28, 2026

Dear Editors,

I am submitting the manuscript “Online Anchor-Robust Forecasting under Economic Regime Shift” for consideration as a research article in *Journal of Forecasting*.

Most online forecasting methods deal with a regime shift after the fact. They detect a change in the loss or in coverage and then recalibrate, so they pay a re-adaptation cost at every break. The paper studies the case in which the shift arrives through observable drivers, such as geopolitical-risk or policy-uncertainty indices, and asks whether a forecaster can be made robust to it in advance.

The method, OARF, is an online version of anchor regression. It penalises the part of the forecast error that is correlated with the drivers, using past-only moving averages of the relevant moments, and it reduces to online gradient descent when the penalty is zero. For the case where the analyst supplies the driver channel, I prove that the learner is no-regret under a notion I call interventional regret, in which the comparator is evaluated under the worst bounded intervention on the drivers. An optional layer estimates the channel from the data. It is unstable on a minority of simulated streams, and the paper proposes a diagnostic, computed without knowledge of the true channel, that flags those streams. Reverting to plain online learning on the flagged streams lowers the layer’s mean worst-case error by about a quarter, although the threshold was set on the same simulations.

The evidence has two parts. On a structural-causal-model benchmark with a held-out grid of interventions, OARF more than halves the median worst-case interventional error of online gradient descent and cuts that of every online baseline by at least 40%, at some cost in ordinary error. Stress tests and assumption-violation scenarios show where this advantage disappears: when the driver is noisy or misspecified, or when regimes shift the target mechanism or the driver’s variance. On sixteen years of daily Brent and Henry Hub data, all methods but one lie in the Model Confidence Set, so OARF is competitive there but not significantly better.

Code, data-build scripts and the commands that regenerate every number, table and figure are available at <https://github.com/vinhqdang/OARF-Online-Anchor-Robust-Forecasting-under-Economic-Regime-Shift>.

The manuscript is original, has not been published, and is not under consideration elsewhere. I have no competing interests to declare. The study uses only public, non-personal data and did not require ethics approval.

Thank you for considering the manuscript.

Sincerely,
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